

Initial Project Planning Template

Date	07 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The backlog below is organized around the project's five delivery epics — data collection, visual analysis, preprocessing, model development, and deployment — each run as a one-week sprint. Team member names and exact dates should be filled in for your cohort; the story points and priorities reflect actual task complexity from the implementation.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Understanding	USN-1	As a developer, I can collect and load the applicant records (application_record.csv) and credit-history records (credit_record.csv) from the Kaggle Credit Card Approval Prediction dataset so I have raw data to work with.	2	High		02 Feb 2026	08 Feb 2026
Sprint-1		USN-2	As a developer, I can design an ER diagram showing how application_record and credit_record link via applicant ID, so the data model is clearly documented.	2	Medium		02 Feb 2026	08 Feb 2026
Sprint-1		USN-3	As a developer, I can perform an initial data-quality check (duplicate applicant IDs, missing values, column types) so downstream steps handle the data correctly.	1	High		02 Feb 2026	08 Feb 2026
Sprint-2	Visualization & Analysis	USN-4	As a developer, I can visualize the distribution of applicant demographic and financial features (income, age, employment, family size) so I understand the applicant population.	2	Medium		09 Feb 2026	15 Feb 2026
Sprint-2		USN-5	As a developer, I can run a vintage (cohort) analysis of credit_record.csv — cumulative bad-rate by months-on-book — to check whether the naïve "ever delinquent" label is reliable at different account ages.	3	High		09 Feb 2026	15 Feb 2026
Sprint-2		USN-6	As a developer, I can visualize the class balance of good vs. risky accounts so I can plan for imbalance-handling during modelling.	1	Medium		09 Feb 2026	15 Feb 2026
Sprint-3	Data Preprocessing	USN-7	As a developer, I can construct the target label using a 24-month minimum months-on-book threshold, so only applicants with enough observed history are labeled, removing label noise from the majority class.	5	High		16 Feb 2026	22 Feb 2026
Sprint-3		USN-8	As a developer, I can engineer features (AGE_YEARS, EMPLOYED_YEARS, NOT_EMPLOYED_FLAG, DEPENDENTS_RATIO, FAMILY_SIZE) from the raw DAYS_BIRTH / DAYS_EMPLOYED / children / family-size fields.	3	High		16 Feb 2026	22 Feb 2026
Sprint-3		USN-9	As a developer, I can encode categorical fields (one-hot for nominal fields, ordinal for education) and cap income outliers at the 99th percentile so the dataset is model-ready.	3	Medium		16 Feb 2026	22 Feb 2026
Sprint-4	Model Development	USN-10	As a developer, I can train and compare baseline models (Logistic Regression, Decision Tree, Random Forest, XGBoost) with default hyperparameters to get an informational performance benchmark.	3	Medium		23 Feb 2026	01 Mar 2026
Sprint-4		USN-11	As a developer, I can run a cross-validated hyperparameter search (RandomizedSearchCV, 40 candidates x 5 folds) for XGBoost, optimizing F1 on the minority reject class.	5	High		23 Feb 2026	01 Mar 2026
Sprint-4		USN-12	As a developer, I can select a probability decision threshold from out-of-fold predictions and validate the final model once on an untouched hold-out test set, so the reported metrics are unbiased.	3	High		23 Feb 2026	01 Mar 2026
Sprint-5	Model Deployment	USN-13	As a user, I can fill in a web form with my personal, income, and employment details to request a credit-approval prediction.	3	High		02 Mar 2026	08 Mar 2026
Sprint-5		USN-14	As a user, I can submit the form and receive an Approve/Reject decision along with the approval probability and the threshold used to reach it.	3	High		02 Mar 2026	08 Mar 2026
Sprint-5		USN-15	As a user, I can see the model's disclosed precision/recall on the result page, so the prediction isn't presented as more reliable than it actually is.	2	Medium		02 Mar 2026	08 Mar 2026